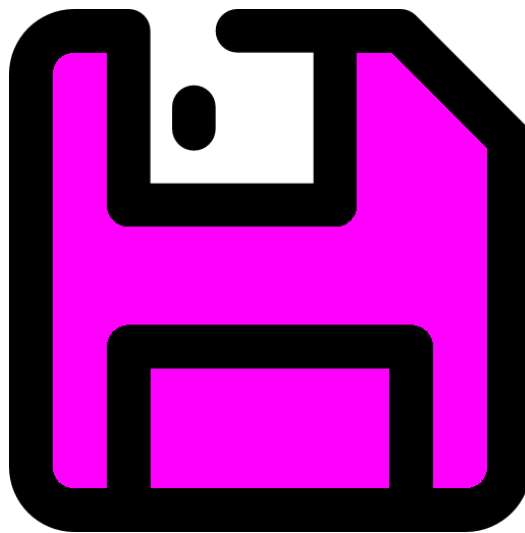


Birds of Impalitism



Unity Save System Documentation



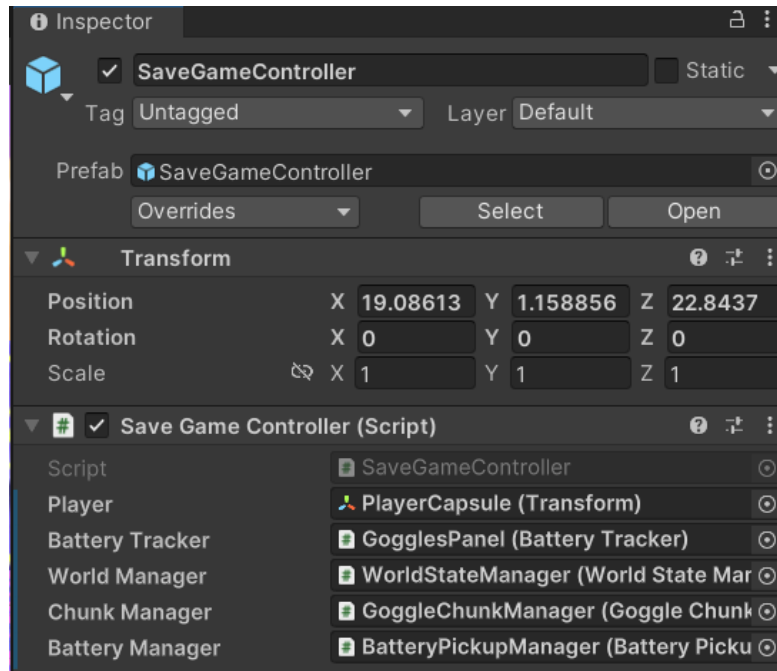
Overview

This document explains how the save system works specifically in Bird of Impalitim. It covers saving and loading player progress, world state, battery pickups, and Google Chunk unlocks across scenes and the Hub world.

Key Components

Path: Assets > Script > SaveSystem > SaveGameController.Prefab

1. SaveGameController.cs



Note - The SaveGameController is not persistent and must be placed in the “Scene Hierarchy.” It may only exist in the Hub world and nowhere else

- Acts as the central save/load handler.
- Stores player position, Battery charge level, world flags, and unlocked goggle chunks.
- Interacts with:
 - **BatteryTracker**
 - **WorldStateManager**
 - **GoggleChunkManager**
 - **BatteryPickupManager**

2. SaveData.cs

Path: Assets > Script > SaveSystem > SaveData.cs

Note - The SaveData should **NOT** be in the "Scene Hierarchy"; it only holds our data and should only be accessed through code.

- Serializable class to hold save fields:
 - **playerX, playerY, playerZ**
 - **playerRotationY**
 - **batteryCharge**
 - **worldFlags**
 - **unlockedChunks**
 - **collectedBatteryIDs**

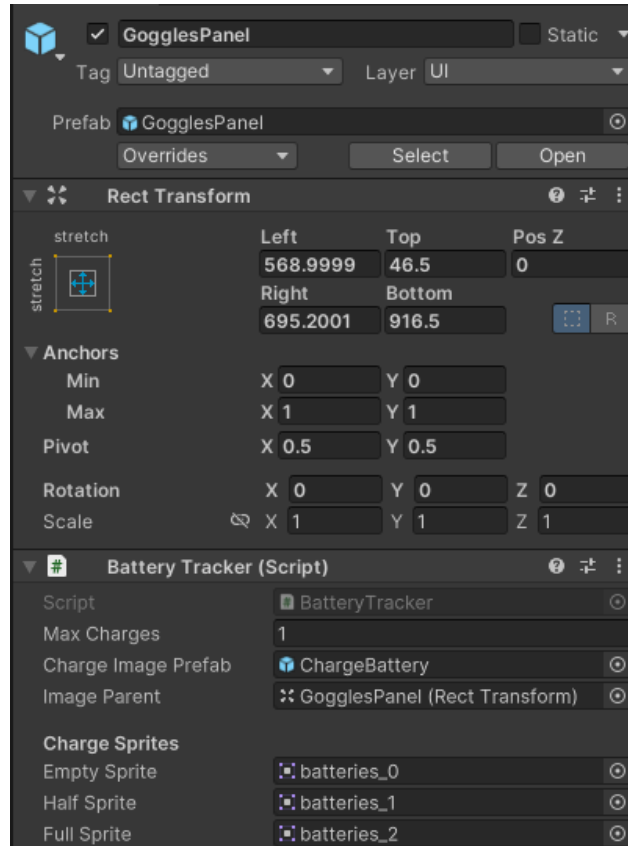
3. SaveSystem.cs

- Handles writing/reading **SaveData** to disk using JSON format.
- Stores in persistent data path.

Battery System

BatteryTracker.cs

Path: Assets > Script > Goggles System > GogglesPanel.Prefab



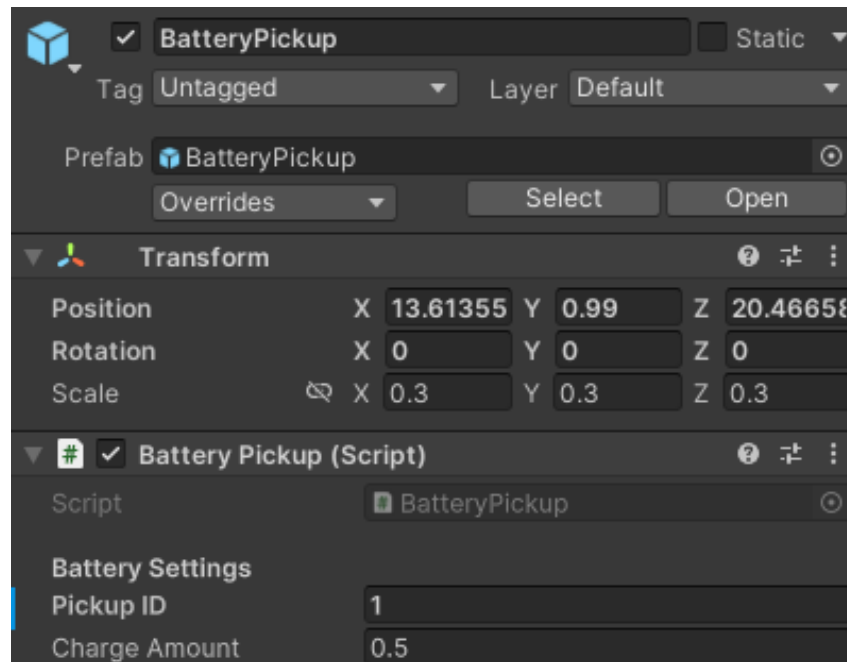
Note - The battery Tracker is attached to “Goggles Panel” prefab under “UICanvas” and must be placed in the “Scene Hierarchy”.

- Tracks the current and max charges.
- Displays them visually with UI Sprites.
- Supports half-charges.
- Methods:

- **AddCharge(float amount)**
- **UseCharge()**
- **GetCurrentChargeLevel()**
- **SetCharge(float amount)**

BatteryPickup.cs

Path: Assets > Script > Goggles System > BatteryPickup.Prefab

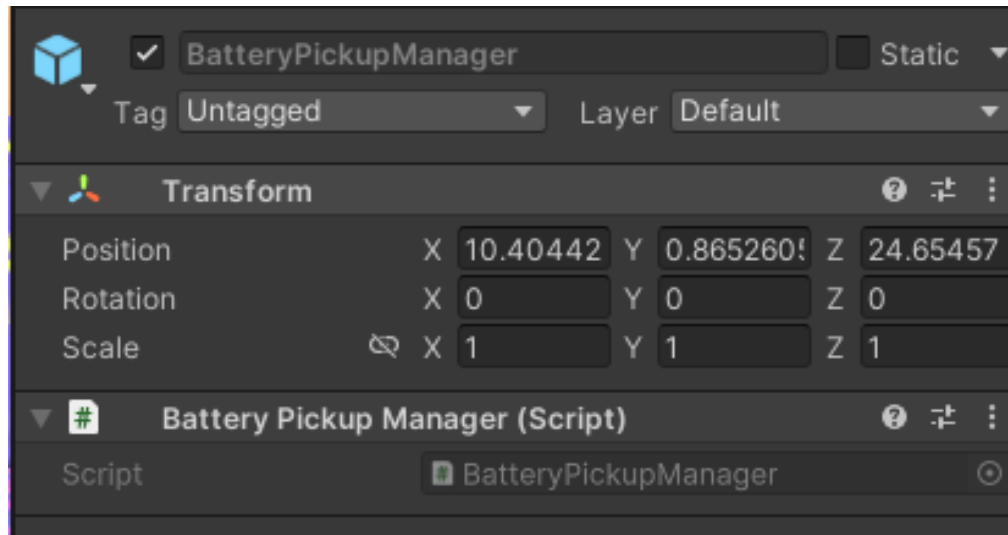


Note - For each BatteryPickup prefab, a unique “Pickup ID” is **MANDATORY** for proper object Save/Loading instances. Must be placed in the “Scene Hierarchy” to work properly

- Detects when the player collects a pickup.
- Adds charge to tracker.
- Communicates with **BatteryPickupManager**.
- Auto disables itself when collected.

BatteryPickupManager.cs

Path: Assets > Script > Goggles System > BatteryPickupManager.Prefab

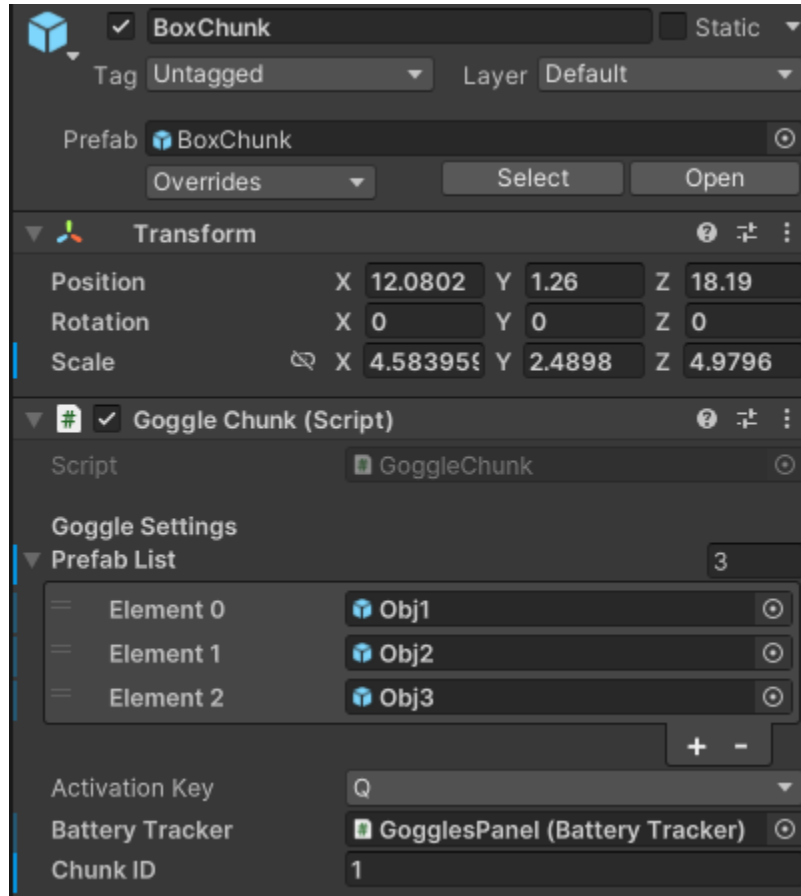


- Maintains collected battery pickup IDs.
- On save: **GetCollectedBatteries()**
- On load: **LoadCollectedBatteries()**
- Re-enables/disables batteries accordingly using **ResetPickupState()**.

Goggle Chunk System

GoggleChunk.cs

Path: Assets > Script > Goggle System > BoxChunk.Prefab

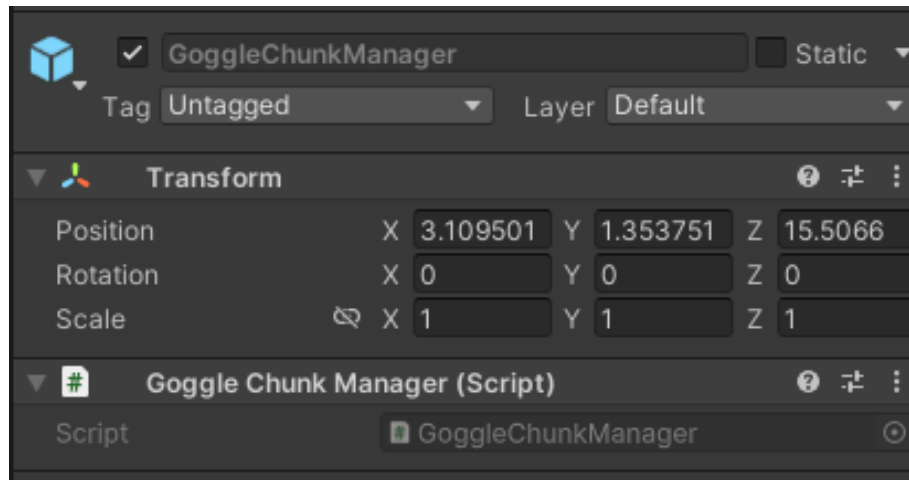


Note - GoggleChunk works just like the BatteryPickup with a unique ID. It is **MANDATORY** to not have a repeating ID. Must be placed in "Scene Hierarchy"

- Modular prefabs that toggle in the world.
- Can only activate with a full charge.
- Once unlocked (first time), no charge needed.

GoggleChunkManager.cs

Path: Assets > Script > Goggles System > GoggleChunkManager.Prefab



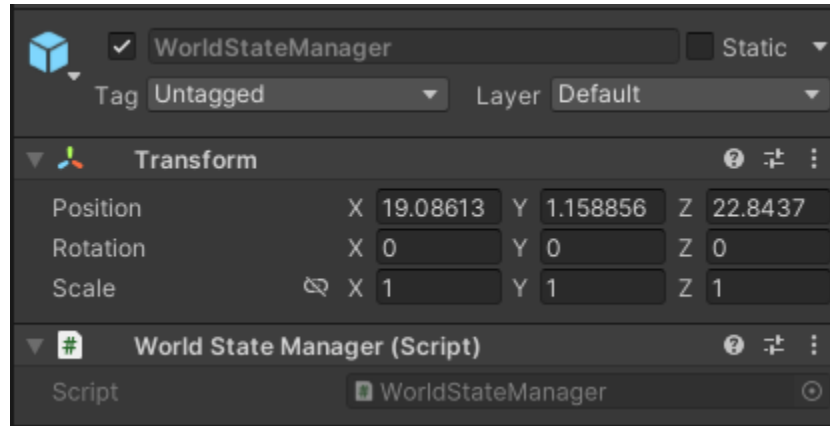
Note - GoggleChunkManager **MUST** be in the “Scene Hierarchy” to properly manage the Goggle Chunks.

- Stores a list of unlocked chunks by ID.
- On load: sets **hasBeenActivated = true**
- Method: **LoadUnlockedChunks()** calls **ResetChunkState()** on each chunk

World Flags

WorldStateManager.cs

Path: Assets > Script > SaveSystem > WorldStateManager.Prefab



Note - WorldstateManager must be in the “Scene Hierarchy” for flagging level completions.

- Tracks boolean flags to represent conditions in the world.
- **SaveFlags()**
- **LoadFlags()**
- Used to persist level completions, doors unlocked, NPC states, etc.

Scene Flow

1. Save Before Scene Change

- Use `SaveGameController.SaveGame()` before leaving the Hub World.

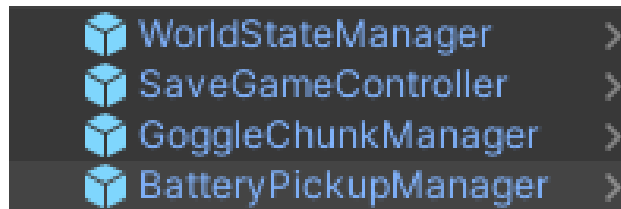
2. Level Completion Unlocks Flag

- Update World Flag in `SaveData` when levels are beaten.

3. Reload Main Scene

- `SaveGameController.Awake()` should call `LoadGame()`
- Load's the player's last Position, Battery UI, goggles, and world states to be restored.

Required Scene Setup



- Add **SaveGameController** Prefab to the “Scene Hierarchy.”
- Assign references to:
 - **Player** (Scene Hierarchy)
 - **BatteryTracker** (Scene Hierarchy)
 - **GoggleChunkManager** (Scene Hierarchy)
 - **BatteryPickupManager** (Scene Hierarchy)
 - **WorldStateManager** (Scene Hierarchy)
- **Mandatory:** all **BatteryPickup** and **GoggleChunk** have unique IDs.

--Continued--